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NONUNIFORM EXPONENTIAL BEHAVIOR VIA EVOLUTION SEMIGROUPS

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Abstract. We provide a unified framework to study the nonuniform exponential behavior of a dynamics using the theory of evolution semigroups. We emphasize that nonuniform hyperbolicity is ubiquitous in the context of ergodic theory: for an autonomous differential equation whose flow preserves a finite measure, the variational equations of almost all trajectories with nonzero Lyapunov exponents have a nonuniform exponential behavior. We consider the general case of a nonautonomous dynamics on a Banach space. In particular, we show that to any nonuniformly exponentially bounded evolution family, one can associate a C_0 semigroup (thus an autonomous dynamics) on some appropriate Banach space. This semigroup is then used to characterize completely the nonuniform hyperbolicity of the evolution family. In addition, we establish a corresponding spectral mapping theorem for an evolution semigroup, which allows us to obtain another characterization of the nonuniform hyperbolicity of an evolution family in terms of the spectrum of the evolution semigroup.

§1. *Introduction.* Our main aim is to provide a unified framework to study the nonuniform exponential behavior of a dynamics using the theory of evolution semigroups. In particular, to any nonuniformly exponentially bounded evolution family we associate a C_0 semigroup on some appropriate Banach space. We then use the semigroup to characterize completely the nonuniform hyperbolicity of the evolution family.

Our work is thus a contribution to the nonuniform hyperbolicity theory. For primary reference sources in the field, we refer the reader to [**1**, **2**]. The study of the hyperbolicity of a dynamics in terms of the spectral properties of certain evolution maps goes back to Mather [**6**]. We refer the reader to [**3**] for references and for a detailed account in the case of uniform hyperbolicity.

1.1. *Evolution semigroups and hyperbolicity.* In order to give a simple illustration of the problems that we have in mind, consider a bounded linear operator A acting on a Banach space X. The asymptotic behavior of the solutions of the linear equation $x' = Ax$ can be completely characterized in terms of the spectrum $\sigma(A)$ of A. Namely, if $\sigma(A)$ does not intersect the imaginary axis, then the equation has an exponential dichotomy. This means that:

(1) there exists a projection $P : X \to X$ satisfying $Pe^{tA} = e^{tA}P$ for $t \in \mathbb{R}$;

(2) there exist $N, \lambda > 0$ such that for $t \geqslant 0$,

$$\|e^{tA}P\| \leqslant Ne^{-\lambda t}, \quad \|e^{-tA}(\mathrm{Id} - P)\| \leqslant Ne^{-\lambda t}.$$

The existence of an exponential dichotomy is obtained via the spectral mapping theorem for a bounded linear operator, which says that

$$\sigma(e^{tA}) = e^{t\sigma(A)} \quad \text{for } t \geqslant 0.$$

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One can also consider an unbounded operator A generating a strongly continuous semigroup e^{tA} for $t \geqslant 0$, but in general it is no longer possible to give a corresponding characterization of the existence of an exponential dichotomy in terms of the spectrum of A due to the absence of a spectral mapping theorem for arbitrary unbounded operators (for various counterexamples, see [4, Section IV.3]). On the other hand, it is possible to give a complete characterization of the hyperbolicity in terms of the spectrum of the generator of the C_0 semigroup T_t defined by

$$(T_t u)(s) = e^{tA} u(s-t)$$

on the set of all continuous functions $u : \mathbb{R} \to X$ vanishing at infinity (see, for example, [3, 4] for the case of a uniform exponential behavior). This characterization depends crucially on the existence of a spectral mapping theorem for the semigroup, even if it may not exist for the original dynamics (as noted above in the autonomous case when the operator A is unbounded).

One can also consider a nonautonomous linear equation

$$x' = A(t)x \tag{1}$$

for some linear operators $A(t)$, for $t \in \mathbb{R}$, acting on a Banach space X (note that the operators may be unbounded). It can be shown that equation (1) is well posed (i.e., we have existence and uniqueness of solutions plus continuous dependence on the initial data) if and only if its solutions are generated by an evolution family $U(t, s)$ (see Section 2 for the definition), in the sense that $x(t) = U(t, s)x(s)$ for $t \geqslant s$ and for any solution x of the equation with initial condition at time s (see [4, Proposition VI.9.3]).

Now let $U(t, s)$ be an evolution family on a Banach space X (not necessarily associated with a nonautonomous linear equation) and assume that it has bounded growth. This means that there exist $M, \alpha > 0$ such that

$$\|U(t, s)x\| \leqslant Me^{\alpha(t-s)}\|x\| \quad \text{for } t \geqslant s,\ x \in X. \tag{2}$$

Then one can define a C_0 semigroup T_t on the set of all continuous functions vanishing at infinity by

$$(T_t u)(s) = U(s, s-t)u(s-t) \tag{3}$$

called the *evolution semigroup* associated with the evolution family. Essentially, it transforms the original nonautonomous problem into an autonomous problem, although at the expense of considering a larger space (which is infinite-dimensional even when X is finite-dimensional). Again one can characterize the hyperbolicity of the evolution family in terms of the generator of the semigroup (see [3, 4]).

We say that the evolution family $U(t, s)$ has a *nonuniform exponential dichotomy* if:

(1) there exist projections $P_t : X \to X$ for $t \in \mathbb{R}$ satisfying

$$P_t U(t, s) = U(t, s)P_s \quad \text{for } t \geqslant s; \tag{4}$$

(2) there exist $\lambda > 0$ and a continuous function $N : \mathbb{R} \to \mathbb{R}^+$ such that

$$\|U(t, s)P_s\| \leqslant N(s)e^{-\lambda(t-s)} \tag{5}$$

and

$$\|U(s, t)Q_t\| \leqslant N(t)e^{-\lambda(t-s)} \tag{6}$$

for $t \geqslant s$, where $Q_t = \mathrm{Id} - P_t$, with the requirement of invertibility of the evolution family in the second inequality.

The main difficulty when trying to use the former approach in the presence of nonuniform hyperbolicity is that the bounded growth condition in (2) may fail, which causes that T_t may

not even be defined, at least on the set of continuous functions vanishing at infinity. We solved this problem by first introducing an appropriate space of functions where T_t becomes a C_0 semigroup.

This can be seen as a starting point to develop a corresponding theory to study nonuniform hyperbolicity and its applications in terms of evolution semigroups. Indeed, having in mind that the theory of semigroups is an important tool in the theory of differential equations (see, for example, [**3**, **9**] for the discussion of many applications), in particular in the presence of uniform hyperbolicity (see [**4**, Section IV.3]), we hope that it may also play a relevant role in the study of nonuniform hyperbolicity (see Section 1.4 for a more detailed discussion).

1.2. *Nonuniform exponential behavior.* We note that a principal motivation to study the nonuniform exponential behavior of a dynamics is that from the point of view of ergodic theory, almost all trajectories with nonzero Lyapunov exponents have a nonuniform exponential behavior. For simplicity of the exposition, consider an autonomous differential equation

$$x' = F(x) \tag{7}$$

for some smooth map $F : \mathbb{R}^n \to \mathbb{R}^n$ on a finite-dimensional space and assume that all solutions are global. The associated flow $\Phi = (\varphi_t)_{t\in\mathbb{R}}$ is defined by $\varphi_t(x_0) = x(t)$, where $x(t)$ is the solution of equation (7) with $x(0) = x_0$. Now assume that Φ preserves a finite measure μ on $\mathbb{R}^n$, that is,

$$\mu(\varphi_t(A)) = \mu(A)$$

for any $t \in \mathbb{R}$ and any measurable set $A \subset \mathbb{R}^n$. For example, any Hamiltonian flow preserves the Liouville measure on each energy level. One can show (see, for example, [**1**]) that for μ-almost all $x \in \mathbb{R}^n$ with Lyapunov exponents

$$\lambda(x, v) = \limsup_{t\to\infty} \frac{1}{t} \log \|d_x\varphi_t v\|$$

nonzero for all $v \neq 0$, the dynamics of the linear variational equation

$$y' = A_x(t)y, \quad \text{where } A_x(t) = d_{\varphi_t(x)}F, \tag{8}$$

is nonuniformly hyperbolic, that is, the equation has a nonuniform exponential dichotomy. This means that for the evolution family $U(t, s)$ associated with equation (8) on $X = \mathbb{R}^n$:

(1) there exist projections $P_t : X \to X$ for $t \in \mathbb{R}$ satisfying (4);
(2) there exist $\lambda, \varepsilon > 0$ such that for $t \geqslant s$,

$$\|U(t, s)P_s\| \leqslant e^{-\lambda(t-s)+\varepsilon|s|}, \quad \|U(s, t)Q_t\| \leqslant e^{-\lambda(t-s)+\varepsilon|s|}, \tag{9}$$

where $Q_t = \mathrm{Id} - P_t$.

In fact, we consider the more general case of strongly continuous evolution families $U(t, s)$ on a Banach space X for which the dynamics may be invertible only along the unstable direction, as well as the more general case of a nonuniform behavior $N(s)$ other than the exponential one $e^{\varepsilon|s|}$ (see (9)). There are also infinite-dimensional results, by Ruelle [**8**] in Hilbert spaces and by Mañé [**5**] in arbitrary Banach spaces, showing that again nonvanishing Lyapunov exponents lead to a nonuniform exponential behavior of the dynamics.

We refer the reader to [**1**] for a detailed exposition of the nonuniform hyperbolicity theory. Among some of its most important developments are the existence of stable and unstable manifolds, their absolute continuity, the ergodic properties of smooth measures, the entropy formula, the ergodic properties of geodesic flows on nonpositively curved manifolds, and the product structure of hyperbolic invariant measures.

1.3. *Description of our results.* As a first step toward a characterization of the nonuniform hyperbolicity of the evolution family in terms of an evolution semigroup, we start by showing that an evolution family is nonuniformly hyperbolic if and only if it is (uniformly) hyperbolic with respect to an appropriate family of norms $\|\cdot\|_t$, in the sense that properties (5) and (6) are replaced, respectively, by

$$\|U(t,s)P_s x\|_t \leqslant Ne^{-\lambda(t-s)}\|x\|_s$$

and

$$\|U(t,s)Q_t x\|_t \leqslant Ne^{-\lambda(t-s)}\|x\|_s,$$

for some constant $N > 0$ and all $t \geqslant s$ and $x \in X$, with the requirement of invertibility of the evolution family in the second inequality. In addition, we show that it is possible to choose the family of norms so that the map $(t,x) \mapsto \|x\|_t$ is continuous on $\mathbb{R} \times X$ (the norms are introduced explicitly at the end of this section). This is the content of Theorem 4, which allows us to study the hyperbolicity of an evolution family with respect to a family of norms instead of its nonuniform hyperbolicity.

After this preliminary reduction, we start developing our construction. In particular, we show that to any nonuniformly exponentially bounded evolution family $U(t,s)$, which means that there exist $\alpha > 0$ and a continuous function $M : \mathbb{R} \to \mathbb{R}^+$ such that

$$\|U(t,s)x\| \leqslant M(s)e^{\alpha(t-s)}\|x\| \quad \text{for } t \geqslant s,\ x \in X, \tag{10}$$

one can associate a C_0 semigroup on an appropriate Banach space. Namely, let C_* be the set of all continuous functions $u : \mathbb{R} \to X$ such that

$$\lim_{|s|\to\infty} \|u(s)\|_s = 0.$$

Then C_* is a Banach space when equipped with the norm

$$\|u\|_* := \sup_{s\in\mathbb{R}} \|u(s)\|_s$$

(see Proposition 5) and we define a linear operator $T_t : C_* \to C_*$ by

$$(T_t u)(s) = U(s, s-t)u(s-t) \quad \text{for } u \in C_*,\ s \in \mathbb{R}.$$

We show that if the evolution family $U(t,s)$ is nonuniformly exponentially bounded, then $\mathcal{T} = (T_t)_{t\geqslant 0}$ is a C_0 semigroup on C_* (see Theorem 6). We call it the *evolution semigroup* associated with the evolution family.

In our first main result (Theorem 7), we characterize completely the nonuniform hyperbolicity of a nonuniformly exponentially bounded evolution family in terms of the evolution semigroup. The following is an equivalent formulation.

THEOREM 1. *A nonuniformly exponentially bounded evolution family is nonuniformly hyperbolic if and only if $\mathcal{T}$ is hyperbolic.*

In addition, in Theorem 8 we establish a spectral mapping theorem for an evolution semigroup in this general context, which allows us to obtain our second main result (Theorem 9): a characterization of the nonuniform hyperbolicity of an evolution family in terms of the spectrum of the evolution semigroup. The following theorem is a partial formulation of the result.

THEOREM 2. *A nonuniformly exponentially bounded evolution family is nonuniformly hyperbolic if and only if*

$$\sigma(T_t) \subset \{z \in \mathbb{C} : |z| \neq 1\}$$

for some $t > 0$ (*and so for all* $t > 0$).

We emphasize that the corresponding results for a uniform exponential behavior (see, for example, [**3**, **4**]) are particular cases of our results in Theorems 1 and 2.

On the other hand, it should be stressed that it is impossible to study a nonuniformly hyperbolic dynamics following the approach in the special case of a uniform exponential behavior. Indeed, the maps T_t introduced in (3) may not be defined on the spaces that are usually considered in the presence of uniform hyperbolicity because condition (2) may fail (such as in Example 1). This causes that one can no longer use spectral mapping theorems via the generator of the semigroup to establish the existence of hyperbolic behavior without further modifications.

In particular, we had to introduce an appropriate Banach space for the phase space of the semigroup. This space is defined in terms of certain Lyapunov norms that did not appear earlier in the literature. Roughly speaking, a Lyapunov norm (or a family of Lyapunov norms $\|\cdot\|_t$) is one for which the nonuniform exponential behavior becomes uniform (see, for example, [**1**]), but it is often the case that it needs to be introduced taking the particular context into account. In our case, we had to consider an asymmetric behavior in the stable and unstable parts (see (17) and (18)) due to the possible nonuniform exponential growth. More precisely, for $x \in X$ and $t \in \mathbb{R}$, we considered the norm (see (17) and (18))

$$\|x\|_t = a(x) + \max\{b(x), c(x)\},$$

where

$$a(x) = \sup_{\tau \geqslant t} \left(\|U(\tau, t)P_t x\| e^{\lambda(\tau - t)}\right),$$

$$b(x) = \sup_{\tau \leqslant t} \left(\|U(\tau, t)Q_t x\| e^{-\lambda(\tau - t)}\right),$$

$$c(x) = \sup_{\tau \geqslant t} \left(\|U(\tau, t)Q_t x\| e^{-\alpha(\tau - t)}\right),$$

with the constants λ and α as in (5), (6), and (10). We emphasize that the usual Lyapunov norms considered in ergodic theory do not require this asymmetry in the stable and unstable parts (see [**1**]). In addition, we showed that the norms $\|x\|_t$ vary continuously with (t, x), which required a special care particularly due to the possible noninvertibility of the evolution family.

1.4. *Further motivation.* We already noted that our work is a contribution to the nonuniform hyperbolicity theory. In this section, we give further motivation for the interest in our results.

First of all, there are two basic aspects that stand out in our approach:

(1) starting with the nonautonomous dynamics given by an evolution family $U(t, s)$, we consider the autonomous dynamics given by an evolution semigroup $\mathcal{T}$;
(2) the nonuniform hyperbolicity of an evolution family $U(t, s)$ is transferred to its uniform hyperbolicity with respect to an appropriate family of norms and then to the uniform hyperbolicity of $\mathcal{T}$.

Although this requires passing to an infinite-dimensional space even more defined in terms of Lyapunov norms, the potential advantage is clear: it is in general simpler to

study an autonomous dynamics and a uniformly hyperbolic dynamics than, respectively, a nonautonomous dynamics and a nonuniformly hyperbolic dynamics. Even more, sometimes it is unclear what should be the appropriate nonautonomous notion corresponding to a given one in the autonomous case (for example, a Morse–Smale system or an equivariant system), while passing to the semigroup $\mathcal{T}$ may help in this respect (because one can transfer a given "autonomous" property for $\mathcal{T}$ to a corresponding property for the original dynamics).

Today the theory of semigroups is an important tool in the theory of differential equations (as well as in many other areas, as illustrated, for example, in [3, 4, 9]). This is not restricted to the study of hyperbolicity, although this is the emphasis in the present paper. Building on the existing theory for a uniform exponential behavior, we give below some pointers for possible future developments and for the potential interest of our work. These developments should require substantial additional work and so must be left for separate publications.

In view of the importance of the notion of hyperbolicity and its consequences in a large part of the theory of differential equations and dynamical systems makes, it is quite relevant to understand whether the exponential behavior persists under sufficiently small linear perturbations. A prototype is the integral equation

$$x(t) = U(t,s)x + \int_s^t U(t,\tau)B(\tau)\,d\tau,$$

which gives the solutions of the differential equation

$$x' = (A(t) + B(t))x$$

when $U(t,s)$ is the evolution family associated with $x' = A(t)x$. This is called the robustness property of the dynamics and has been studied in many situations. In the context of our work, this problem is transferred to that of the hyperbolicity of the generator of the semigroup, which may lead to somewhat natural and more direct developments using the existing results on the perturbations of the spectrum of a linear operator.

In another direction, building on the Grobman–Hartman theorem in a neighborhood of a hyperbolic fixed point, there are nonautonomous versions of the theorem. These are essentially trivial in the case of discrete time when there is no requirement on the conjugacies. Indeed, the solution of a system of equations

$$h_{n+1} \circ f_n = f_{n+1} \circ h_n$$

for some invertible maps f_n, for $n \in \mathbb{N}$, can be obtained setting $h_1 = \mathrm{Id}$ and then iterating to obtain the conjugacies in the form

$$h_n = f_n \circ \cdots \circ f_2 \circ f_1^{-1} \circ \cdots \circ f_{n-1}^{-1}.$$

The situation is much more complicated when there is a requirement on the conjugacies, such as boundedness, and even more for continuous time. Again the theory of semigroups may be of help in this respect, with the topological conjugacies induced by a single corresponding conjugacy for the semigroup.

Still another possible application concerns the Sacker–Sell spectral theory. In view of our spectral mapping theorem relating the spectra of the semigroup and its generator, one can study Sacker–Sell spectra by considering only the generator, and study properties such as the continuous dependence on the Hausdorff distance under perturbations. Since we are dealing with nonuniform hyperbolicity, it is natural to consider this approach for the sharper nonuniform spectrum introduced recently (it is defined replacing the hyperbolicity by nonuniform hyperbolicity in the usual spectrum).

Finally, in some situation, it is unavoidable to leave the ambient space. This is the case, for example, of delay equations, where the dynamics naturally lives in an infinite-dimensional space. While it is well known that the existence of a hyperbolic behavior for an autonomous linear delay equation can be characterized in terms of the generator of the dynamics on the infinite-dimensional space, to our knowledge there are no corresponding results in the case of a nonautonomous linear delay equation. We believe that the theory of semigroups may be of help in this respect.

The theory of semigroups should be of help in all these situations, for example, making many arguments simpler (since we are considering a uniformly hyperbolic autonomous dynamics) as well as in view of obtaining nontrivial information about a dynamics in an easier way. As noted above, it may also lead the way to find some appropriate "nonautonomous" notions when they are not available yet, or to provide simpler proofs of existing results. Another principal advantage is that the theory of semigroups is amenable to consider perturbations described in terms of the spectra of the generators (up to the well-posedness of the equations, which is a different problem).

§2. *Nonuniformly hyperbolic evolution families.* In this section, we show that the nonuniformly hyperbolic evolution families can be characterized in terms of their *uniform* hyperbolicity with respect to a certain family of norms.

Let $X = (X, \|\cdot\|)$ be a Banach space and let $\mathcal{L}(X)$ be the set of all bounded linear operators acting on X. A (*strongly continuous*) *evolution family* on X is a family $\mathcal{U} = (U(t,s))_{t,s\in\mathbb{R},t\geqslant s}$ of linear operators in $\mathcal{L}(X)$ such that:

(1) $U(t,t) = \mathrm{Id}$ and $U(t,s)\,U(s,r) = U(t,r)$ for $t \geqslant s \geqslant r$;
(2) the map $(t,s) \mapsto U(t,s)\,x$ is continuous for each $x \in X$.

Now let $\mathcal{U}$ be an evolution family on X and write $\mathbb{R}^+ = (0, +\infty)$.

Definition 1. We say that $\mathcal{U}$ is *nonuniformly exponentially bounded* if there exist $\alpha > 0$ and a continuous function $M : \mathbb{R} \to \mathbb{R}^+$ such that

$$\|U(t,s)x\| \leqslant M(s)e^{\alpha(t-s)}\|x\| \quad \text{for } t \geqslant s,\ x \in X.$$

We say that $\mathcal{U}$ is *nonuniformly hyperbolic* (or that it has a *nonuniform exponential dichotomy*) if:

(1) there exist projections $P_t : X \to X$ for $t \in \mathbb{R}$ satisfying

$$P_t U(t,s) = U(t,s)P_s \quad \text{for } t \geqslant s; \tag{11}$$

(2) the map $U(t,s)|_{Q_s(X)} : Q_s(X) \to Q_t(X)$, where $Q_s = \mathrm{Id} - P_s$, is onto and invertible for each $t \geqslant s$;
(3) there exist $\lambda > 0$ and a continuous function $N : \mathbb{R} \to \mathbb{R}^+$ such that

$$\|U(t,s)P_s x\| \leqslant N(s)e^{-\lambda(t-s)}\|x\| \tag{12}$$

and

$$\|V(s,t)Q_t x\| \leqslant N(t)e^{-\lambda(t-s)}\|x\| \tag{13}$$

for $t \geqslant s$ and $x \in X$, where $V(s,t) = (U(t,s)|_{Q_s(X)})^{-1}$.

Taking $t = s$ in (12) and (13), we find that

$$\|P_t\| \leqslant N(t) \quad \text{and} \quad \|P_t\| \leqslant N(t).$$

Hence, $P_t, Q_t \in \mathcal{L}(X)$. We also say that $\mathcal{U}$ is *uniformly hyperbolic* (or that it has a *uniform exponential dichotomy*) if it is nonuniformly hyperbolic with a constant function N.

It turns out that the map $t \mapsto P_t$ is automatically strongly continuous. In fact, we have the following stronger result.

PROPOSITION 3. *Let $\mathcal{U}$ be a nonuniformly hyperbolic evolution family on the Banach space X. Then the map $(t, x) \mapsto P_t x$ is continuous on $\mathbb{R} \times X$.*

Proof. First we establish the continuity of the map $t \mapsto P_t x$ for each $x \in X$. For $t \geqslant s$, we have

$$\begin{aligned}\|P_t x - P_s x\| &\leqslant \|P_t x - P_t U(t,s)x\| + \|U(t,s)P_s x - P_s x\| \\ &\leqslant N(t)\|x - U(t,s)x\| + \|U(t,s)P_s x - P_s x\|.\end{aligned}$$

Hence, letting $t \searrow s$ we find that the map $t \mapsto P_t x$ is continuous from the right at s, and so at all points of $\mathbb{R}$ (since s is arbitrary).

For the continuity from the left, first observe that for each $z \in Q_t(X)$ the map

$$(-\infty, t] \ni s \mapsto V(s,t)z = \big(U(t,s)|_{Q_s(X)}\big)^{-1} z$$

is continuous. Indeed, taking $r < s \leqslant t$ and $y \in Q_r(X)$ with $U(t,r)y = z$, we obtain

$$\lim_{\tau \to s} \|V(\tau,t)z - V(s,t)z\| = \lim_{\tau \to s} \|U(\tau,r)y - U(s,r)y\| = 0$$

when $s < t$ and

$$\lim_{\tau \nearrow t} \|V(\tau,t)z - z\| = \lim_{\tau \to t} \|U(\tau,r)y - U(t,r)y\| = 0$$

when $s = t$, since the map $\tau \mapsto U(\tau,r)y$ is continuous. Hence, for $t \geqslant s$ we have

$$\begin{aligned}\|P_t x - P_s x\| = \|Q_s x - Q_t x\| &= \|V(s,t)Q_t U(t,s)x - Q_t x\| \\ &\leqslant \|V(s,t)Q_t[U(t,s)x - x]\| + \|V(s,t)Q_t x - Q_t x\| \\ &\leqslant N(t)e^{-\lambda(t-s)}\|U(t,s)x - x\| + \|V(s,t)z - z\|,\end{aligned}$$

where $z = Q_t x$. Letting $s \nearrow t$ this implies that the map $t \mapsto P_t x$ is continuous from the left at t, and so at all points of $\mathbb{R}$, since both maps $s \mapsto U(t,s)x$ and $s \mapsto V(s,t)z$ are continuous for $s \leqslant t$.

Finally, we establish the joint continuity of the map $(t, x) \mapsto P_t x$. Given $t, s \in \mathbb{R}$ and $x, y \in X$, we have

$$\begin{aligned}\|P_t x - P_s y\| &\leqslant \|P_t x - P_s x\| + \|P_s x - P_s y\| \\ &\leqslant \|P_t x - P_s x\| + N(s)\|x - y\|.\end{aligned} \tag{14}$$

Now let $(s, y) \to (t, x)$. It follows from the continuity in the first component that $P_s y \to P_t x$. Moreover, since N is continuous and so bounded in a neighborhood of t, it follows from (14) that $P_s y \to P_t x$ when $(s, y) \to (t, x)$. This completes the proof of the proposition. □

For the next definition, let $\|\cdot\|_t$ for $t \in \mathbb{R}$ be a family of norms on X.

Definition 2. We say that $\mathcal{U}$ is (*uniformly*) *exponentially bounded with respect to the norms* $\|\cdot\|_t$ if there exist $\alpha, M > 0$ such that

$$\|U(t,s)x\|_t \leqslant Me^{\alpha(t-s)}\|x\|_s \quad \text{for } t \geqslant s,\ x \in X. \tag{15}$$

We say that $\mathcal{U}$ is (*uniformly*) *hyperbolic* or that it has a (*uniform*) *exponential dichotomy with respect to the norms* $\|\cdot\|_t$ if:

(1) conditions 1 and 2 in Definition 1 hold;

(2) there exist $\lambda, N > 0$ such that

$$\|U(t,s)P_s x\|_t \leqslant Ne^{-\lambda(t-s)}\|x\|_s \quad \text{and} \quad \|V(s,t)Q_t x\|_s \leqslant Ne^{-\lambda(t-s)}\|x\|_t$$

for $t \geqslant s$ and $x \in X$, where $V(s,t) = (U(t,s)|_{Q_s(X)})^{-1}$.

The following result gives a characterization of the nonuniform hyperbolicity of an evolution family in terms of its (uniform) hyperbolicity with respect to an appropriate family of norms.

THEOREM 4. *Let $\mathcal{U}$ be an evolution family on a Banach space X. Then the following properties are equivalent:*

(1) *$\mathcal{U}$ is nonuniformly exponentially bounded and nonuniformly hyperbolic;*

(2) *$\mathcal{U}$ is exponentially bounded and hyperbolic with respect to some norms $\|\cdot\|_t$ satisfying*

$$\|x\| \leqslant \|x\|_t \leqslant R(t)\|x\| \quad \text{for } t \in \mathbb{R},\ x \in X \tag{16}$$

and some continuous function $R : \mathbb{R} \to \mathbb{R}$, with the map $(t,x) \mapsto \|x\|_t$ continuous on $\mathbb{R} \times X$.

Proof. ($2 \Rightarrow 1$) We first assume that the evolution family $\mathcal{U}$ is exponentially bounded and hyperbolic with respect to some norms $\|\cdot\|_t$ as in property 2. Then

$$\|U(t,s)P_s x\| \leqslant \|U(t,s)P_s x\|_t \leqslant Ne^{-\lambda(t-s)}\|x\|_s \leqslant R(s)Ne^{-\lambda(t-s)}\|x\|$$

and

$$\|V(s,t)Q_t x\| \leqslant \|V(s,t)Q_t x\|_s \leqslant Ne^{-\lambda(t-s)}\|x\|_t \leqslant R(t)Ne^{-\lambda(t-s)}\|x\|$$

for $t \geqslant s$ and $x \in X$. This shows that $\mathcal{U}$ is nonuniformly hyperbolic. Furthermore, for $t \geqslant s$, we have

$$\begin{aligned}\|U(t,s)x\| &\leqslant \|U(t,s)P_s x\|_t + \|U(t,s)Q_s x\|_t \\ &\leqslant \left(Ne^{-\lambda(t-s)} + NMe^{\alpha(t-s)}\right)\|x\|_s \\ &\leqslant R(s)(N+NM)e^{\alpha(t-s)}\|x\|\end{aligned}$$

and so the evolution family $\mathcal{U}$ is nonuniformly exponentially bounded.

($1 \Rightarrow 2$) Conversely, assume that the evolution family $\mathcal{U}$ is nonuniformly exponentially bounded and nonuniformly hyperbolic. Without loss of generality, we assume that $\alpha \geqslant \lambda$. We divide the proof into steps.

Step 1. *Construction of a family of norms.* For $x \in X$ and $t \in \mathbb{R}$, we consider the norm

$$\|x\|_t = \|P_t x\|_t + \|Q_t x\|_t,$$

where

$$\|y\|_t = \sup_{\tau \geqslant t}\left(\|U(\tau,t)y\|e^{\lambda(\tau-t)}\right) \tag{17}$$

for $y \in P_t(X)$ and

$$\|z\|_t = \max\left\{\sup_{\tau \leqslant t}\left(\|V(\tau,t)z\|e^{-\lambda(\tau-t)}\right), \sup_{\tau \geqslant t}\left(\|U(\tau,t)z\|e^{-\alpha(\tau-t)}\right)\right\} \tag{18}$$

for $z \in Q_t(X)$. Note that

$$\|y\|_t \geqslant \|y\| \quad \text{and} \quad \|z\|_t \geqslant \|z\|$$

and so

$$\|x\|_t = \|P_t x\|_t + \|Q_t x\|_t \geqslant \|P_t x\| + \|Q_t x\| \geqslant \|x\|.$$

Moreover,

$$\|P_t x\|_t \leqslant N(t)\|x\| \quad \text{and} \quad \|Q_t x\|_t \leqslant \max\{N(t), M(t)\}\|x\|$$

and so

$$\|x\|_t = \|P_t x\|_t + \|Q_t x\|_t \leqslant (N(t) + \max\{N(t), M(t)\})\|x\|.$$

This shows that the norms $\|\cdot\|_t$ satisfy (16) with the continuous function

$$R(t) = N(t) + \max\{N(t), M(t)\}$$

(since the functions M and N are continuous).

Step 2. *Hyperbolicity of* $\mathcal{U}$. We have

$$\begin{aligned} \|U(t,s)P_s x\|_t &= \sup_{\tau \geqslant t} \left(\|U(\tau,t)U(t,s)P_s x\| e^{\lambda(\tau-t)}\right) = e^{\lambda(s-t)} \sup_{\tau \geqslant t} \left(\|U(\tau,s)P_s x\| e^{\lambda(\tau-s)}\right) \\ &\leqslant e^{\lambda(s-t)} \sup_{\tau \geqslant s} \left(\|U(\tau,s)P_s x\| e^{\lambda(\tau-s)}\right) \leqslant e^{\lambda(s-t)} \|P_s x\|_s \leqslant e^{\lambda(s-t)} \|x\|_s \end{aligned} \tag{19}$$

for $t \geqslant s$ and

$$\begin{aligned} \|V(t,s)Q_s x\|_t &= \max\left\{ \sup_{\tau \leqslant t} \left(\|V(\tau,t)V(t,s)Q_s x\| e^{-\lambda(\tau-t)}\right), \right. \\ &\qquad \left. \sup_{\tau \geqslant t} \left(\|U(\tau,t)V(t,s)Q_s x\| e^{-\alpha(\tau-t)}\right) \right\} \\ &\leqslant \max\left\{ \sup_{\tau \leqslant t} \left(\|V(\tau,s)Q_s x\| e^{-\lambda(\tau-t)}\right), \sup_{s \geqslant \tau \geqslant t} \left(\|V(\tau,s)Q_s x\| e^{-\alpha(\tau-t)}\right), \right. \\ &\qquad \left. \sup_{\tau \geqslant s} \left(\|U(\tau,s)Q_s x\| e^{-\alpha(\tau-t)}\right) \right\} \\ &\leqslant \max\left\{ \sup_{\tau \leqslant s} \left(\|V(\tau,s)Q_s x\| e^{-\lambda(\tau-t)}\right), \sup_{\tau \geqslant s} \left(\|U(\tau,s)Q_s x\| e^{-\alpha(\tau-t)}\right) \right\} \\ &= \max\left\{ e^{-\lambda(s-t)} \sup_{\tau \leqslant s} \left(\|V(\tau,s)Q_s x\| e^{-\lambda(\tau-s)}\right), e^{-\alpha(s-t)} \sup_{\tau \geqslant s} \left(\|U(\tau,s)Q_s x\| e^{-\alpha(\tau-s)}\right) \right\} \\ &\leqslant e^{-\lambda(s-t)} \|Q_s x\|_s \leqslant e^{-\lambda(s-t)} \|x\|_s \end{aligned}$$

for $t \leqslant s$, since $\alpha \geqslant \lambda$. Hence, $\mathcal{U}$ is hyperbolic with respect to the norms $\|\cdot\|_t$.

Step 3. *Exponential boundedness of* $\mathcal{U}$. Now we prove that $\mathcal{U}$ is exponentially bounded with respect to these norms. To this end, note that for $t \geqslant s$, we have

$$\begin{aligned}
\|U(t,s)Q_s x\|_t &= \max\left\{\sup_{\tau\leqslant t}\left(\|V(\tau,t)U(t,s)Q_s x\|e^{-\lambda(\tau-t)}\right),\right.\\
&\quad\left.\sup_{\tau\geqslant t}\left(\|U(\tau,t)U(t,s)Q_s x\|e^{-\alpha(\tau-t)}\right)\right\}\\
&\leqslant \max\left\{\sup_{\tau\leqslant s}\left(\|V(\tau,s)Q_s x\|e^{-\lambda(\tau-t)}\right),\ \sup_{s\leqslant\tau\leqslant t}\left(\|U(\tau,s)Q_s x\|e^{-\alpha(\tau-t)}\right),\right.\\
&\quad\left.\sup_{\tau\geqslant t}\left(\|U(\tau,s)Q_s x\|e^{-\alpha(\tau-t)}\right)\right\} \qquad (20)\\
&\leqslant \max\left\{\sup_{\tau\leqslant s}\left(\|V(\tau,s)Q_s x\|e^{-\lambda(\tau-t)}\right),\ \sup_{\tau\geqslant s}\left(\|U(\tau,s)Q_s x\|e^{-\alpha(\tau-t)}\right)\right\}\\
&= \max\left\{e^{\lambda(t-s)}\sup_{\tau\leqslant s}\left(\|V(\tau,s)Q_s x\|e^{-\lambda(\tau-s)}\right),\ e^{\alpha(t-s)}\sup_{\tau\geqslant s}\left(\|U(\tau,s)Q_s x\|e^{-\alpha(\tau-s)}\right)\right\}\\
&\leqslant e^{\alpha(t-s)}\|Q_s x\|_s \leqslant e^{\alpha(t-s)}\|x\|_s.
\end{aligned}$$

Using (19) and (20), for $t \geqslant s$ we obtain

$$\begin{aligned}
\|U(t,s)x\|_t &= \|U(t,s)P_s x\|_t + \|U(t,s)Q_s x\|_t\\
&\leqslant e^{\lambda(s-t)}\|P_s x\|_s + e^{\alpha(t-s)}\|Q_s x\|_s \leqslant e^{\alpha(t-s)}\|x\|_s,
\end{aligned}$$

which shows that $\mathcal{U}$ is exponentially bounded with respect to the norms $\|\cdot\|_t$.

Step 4. *Continuity of the norms.* It remains to show that the map $(t,x)\mapsto\|x\|_t$ is continuous on $\mathbb{R}\times X$. First, we establish the continuity in t. Take $x\in X$. Since

$$\begin{aligned}
\|x\|_t &= \sup_{\tau\geqslant t}\left(\|U(\tau,t)P_t x\|e^{\lambda(\tau-t)}\right)\\
&\quad+\max\left\{\sup_{\tau\leqslant t}\left(\|V(\tau,t)Q_t x\|e^{-\lambda(\tau-t)}\right),\ \sup_{\tau\geqslant t}\left(\|U(\tau,t)Q_t x\|e^{-\alpha(\tau-t)}\right)\right\},
\end{aligned} \qquad (21)$$

it suffices to prove that each supremum in (21) varies continuously with t. The argument is analogous for all of them and so we consider only the one that is potentially more complicated, that is, the function

$$F(t) = \sup_{\tau\geqslant t}\left(\|U(\tau,t)P_t x\|e^{\lambda(\tau-t)}\right),$$

since the evolution family need not be invertible on the spaces $P_t(X)$. Take $\varepsilon>0$. Then

$$F(t+\varepsilon) = \sup_{\tau\geqslant t+\varepsilon}\left(\|U(\tau,t+\varepsilon)P_{t+\varepsilon}x\|e^{\lambda(\tau-t-\varepsilon)}\right).$$

Since $\varepsilon\mapsto U(t+\varepsilon,t)x$ is continuous, given $\delta>0$ we have

$$\|U(t+\varepsilon,t)x - x\| < \delta$$

for any sufficiently small $\varepsilon > 0$. Therefore,

$$\begin{aligned}\|U(\tau,t)P_t x\| &= \|U(\tau,t+\varepsilon)P_{t+\varepsilon}U(t+\varepsilon,t)x\|\\ &\leqslant \|U(\tau,t+\varepsilon)P_{t+\varepsilon}x\| + \|U(\tau,t+\varepsilon)P_{t+\varepsilon}\|\delta\end{aligned}$$

and

$$\begin{aligned}&\sup_{\tau\geqslant t+\varepsilon}\left(\|U(\tau,t)P_t x\|e^{\lambda(\tau-t)}\right)\\ &\leqslant \sup_{\tau\geqslant t+\varepsilon}\left(\|U(\tau,t+\varepsilon)P_{t+\varepsilon}x\|e^{\lambda(\tau-t)}\right) + \sup_{\tau\geqslant t+\varepsilon}\left(\|U(\tau,t+\varepsilon)P_{t+\varepsilon}\|e^{\lambda(\tau-t)}\right)\delta\\ &\leqslant e^{\lambda\varepsilon}F(t+\varepsilon) + N(t+\varepsilon)e^{\lambda\varepsilon}\delta.\end{aligned}$$

Hence,

$$F(t) \leqslant \max\left\{\sup_{\tau\in[t,t+\varepsilon]}\left(\|U(\tau,t)P_t x\|e^{\lambda(\tau-t)}\right), e^{\lambda\varepsilon}F(t+\varepsilon) + N(t+\varepsilon)e^{\lambda\varepsilon}\delta\right\}.$$

Since the function N is continuous, letting $\varepsilon \searrow 0$ we obtain

$$F(t) \leqslant \max\left\{\|P_t x\|, \liminf_{\varepsilon\searrow 0} F(t+\varepsilon) + N(t)\delta\right\}$$

and the arbitrariness of δ yields that

$$F(t) \leqslant \max\left\{\|P_t x\|, \liminf_{\varepsilon\searrow 0} F(t+\varepsilon)\right\}. \tag{22}$$

In view of Proposition 3, we have

$$\liminf_{\varepsilon\searrow 0} F(t+\varepsilon) \geqslant \liminf_{\varepsilon\searrow 0}\|P_{t+\varepsilon}x\| = \|P_t x\|,$$

and so, by (22),

$$F(t) \leqslant \liminf_{\varepsilon\searrow 0} F(t+\varepsilon). \tag{23}$$

Similarly, we have

$$\begin{aligned}\|U(\tau,t)P_t x\| &= \|U(\tau,t+\varepsilon)P_{t+\varepsilon}U(t+\varepsilon,t)x\|\\ &\geqslant \|U(\tau,t+\varepsilon)P_{t+\varepsilon}x\| - \|U(\tau,t+\varepsilon)P_{t+\varepsilon}\|\delta\end{aligned}$$

and

$$F(t) \geqslant \sup_{\tau\geqslant t+\varepsilon}\left(\|U(\tau,t)P_t x\|e^{\lambda(\tau-t)}\right) \geqslant e^{\lambda\varepsilon}F(t+\varepsilon) - N(t+\varepsilon)e^{\lambda\varepsilon}\delta.$$

Hence,

$$F(t) \geqslant \limsup_{\varepsilon\searrow 0} F(t+\varepsilon) - N(t)\delta$$

and the arbitrariness of δ yields that

$$F(t) \geqslant \limsup_{\varepsilon\searrow 0} F(t+\varepsilon).$$

Together with (23) this implies that $F(t^+) = F(t)$. One can show in a similar manner that $F(t^-) = F(t)$, thus establishing the continuity of F.

Finally, we establish the joint continuity of the map $(t, x) \mapsto \|x\|_t$. Given $t, s \in \mathbb{R}$ and $x, y \in X$, we have

$$|\|x\|_s - \|y\|_s| \leqslant \|x - y\|_s \leqslant R(s)\|x - y\|$$

and so

$$\begin{aligned} |\|x\|_t - \|y\|_s| &\leqslant |\|x\|_t - \|x\|_s| + |\|x\|_s - \|y\|_s| \\ &\leqslant |\|x\|_t - \|x\|_s| + R(s)\|x - y\|. \end{aligned} \tag{24}$$

Now let $(s, y) \to (t, x)$. It follows from the continuity in the first component established before that $\|x\|_s \to \|x\|_t$. Moreover, since R is continuous and so bounded in a neighborhood of t, it follows from (24) that $\|y\|_s \to \|x\|_t$ when $(s, y) \to (t, x)$. This completes the proof of the theorem. □

§3. *C_0 semigroups and families of norms.* In this section, we show that to any nonuniformly exponentially bounded evolution family one can associate a C_0 semigroup on some appropriate Banach space. This semigroup will be used in Section 4 to characterize completely the nonuniform hyperbolicity of the evolution family.

3.1. *Construction of a Banach space.* Let X be a Banach space and consider a family of norms $\|\cdot\|_t$ for $t \in \mathbb{R}$ on X. We shall always assume from now on in the paper that (16) holds for some continuous function $R : \mathbb{R} \to \mathbb{R}^+$ and that the map $(t, x) \mapsto \|x\|_t$ is continuous on $\mathbb{R} \times X$. Let

$$C_* = \left\{ u \in C(X) : \lim_{|s| \to \infty} \|u(s)\|_s = 0 \right\}, \tag{25}$$

where $C(X)$ is the set of all continuous functions $u : \mathbb{R} \to X$, and define

$$\|u\|_* = \sup_{s \in \mathbb{R}} \|u(s)\|_s$$

for $u \in C_*$. It is easy to verify that $\|\cdot\|_*$ is a norm on C_* and that $\|u\|_* < \infty$ for each $u \in C_*$.

PROPOSITION 5. *$(C_*, \|\cdot\|_*)$ is a Banach space.*

Proof. Let $(u_n)_{n \in \mathbb{N}} \subset C_*$ be a Cauchy sequence. Given $\varepsilon > 0$, there exists $p \in \mathbb{N}$ such that

$$\sup_{s \in \mathbb{R}} \|u_n(s) - u_m(s)\|_s < \varepsilon \tag{26}$$

for $n, m > p$. In view of (16), this implies that $(u_n)_{n \in \mathbb{N}}$ is also a Cauchy sequence in $C(X)$ with respect to the supremum norm $\|\cdot\|_\infty$. Hence, u_n converges in $C(X)$ to some function $u \in C(X)$. Again in view of (16), we have

$$\|u_n(s) - u(s)\|_s \leqslant R(s)\|u_n(s) - u(s)\| \leqslant R(s)\|u_n - u\|_\infty \to 0$$

when $n \to \infty$. On the other hand, it follows from (26) that

$$\|u_n(s) - u_m(s)\|_s < \varepsilon \quad \text{for } n, m > p$$

and so

$$\|u_n(s) - u(s)\|_s \leqslant \varepsilon \quad \text{for } n > p.$$

This shows that $\|u_n - u\|_* \to 0$ when $n \to \infty$. Finally, we have

$$\|u_n(s)\|_s - \varepsilon \leqslant \|u(s)\|_s \leqslant \|u_n(s)\|_s + \varepsilon$$

and letting $|s| \to \infty$ yields that

$$\varepsilon \leqslant \liminf_{|s|\to\infty} \|u(s)\|_s \leqslant \limsup_{|s|\to\infty} \|u(s)\|_s \leqslant \varepsilon.$$

The arbitrariness of ε implies that $u \in C_*$. □

Let also $C_c(X) \subset C(X)$ be the set of all continuous functions with compact support. Clearly, $C_c(X)$ is a dense subspace of C_*.

We give an example of an evolution family that is nonuniformly exponentially bounded and nonuniformly hyperbolic, but not uniformly exponentially bounded neither uniformly hyperbolic both with respect to the original norm $\|\cdot\|$ (i.e., with respect to the norms $\|\cdot\|'_t = \|\cdot\|$ for $t \in \mathbb{R}$).

Example 1. Take $a > 1$. The evolution family associated with the differential equation

$$x' = (-a + \sin t + t\cos t)x, \quad t \in \mathbb{R},$$

is $U(t,s) = e^{-a(t-s)+t\sin t - s\sin s}$. Taking $P_s = \mathrm{Id}$, we have

$$\begin{aligned}\|U(t,s)P_s\| &\leqslant e^{(-a+1)(t-s)} e^{(t-s)(\sin t - 1)+s\sin t - s\sin s}\\ &\leqslant e^{(-a+1)(t-s)} e^{|s|-s\sin s}\end{aligned}$$

for $t \geqslant s$. This shows that $\mathcal{U}$ is nonuniformly hyperbolic. Now note that if

$$\|U(t,s)P_s\| \leqslant Ne^{-\lambda(t-s)}$$

for some $N, \lambda > 0$ and all $t \geqslant s$, then taking $s = 2n\pi$ and $t = 2n\pi + \pi/2$ with $n \in \mathbb{Z}$, we obtain

$$e^{-a\pi/2+2n\pi+\pi/2} \leqslant Ne^{-\lambda\pi/2}. \tag{27}$$

But inequality (27) fails for all sufficiently large n. Hence, $\mathcal{U}$ is not uniformly hyperbolic. On the other hand,

$$\begin{aligned}\|U(t,s)P_s x\|_s &= \sup_{\tau\geqslant t} \left(\|U(\tau,t)U(t,s)P_s x\| e^{(a-1)(\tau-t)}\right)\\ &\leqslant e^{-(a-1)(t-s)} \sup_{\tau\geqslant s} \left(\|U(\tau,s)P_s x\| e^{(a-1)(\tau-s)}\right)\\ &= e^{-(a-1)(t-s)}\|x\|_s.\end{aligned}$$

3.2. *Construction of a C_0 semigroup.* A *semigroup* on a Banach space Y is a family $\mathcal{T} = (T_t)_{t\geqslant 0}$ of linear operators in $\mathcal{L}(Y)$ such that

$$T_0 = \mathrm{Id} \quad \text{and} \quad T_t T_s = T_{t+s} \quad \text{for } t, s \geqslant 0.$$

It is called a C_0 *semigroup* or a *strongly continuous semigroup* if

$$\lim_{t\searrow 0} T_t u = u \quad \text{for } u \in Y.$$

Any C_0 semigroup $\mathcal{T}$ is *exponentially bounded*, that is, there exist $\alpha, M > 0$ such that

$$\|T_t\| \leqslant Me^{\alpha t} \quad \text{for } t \geqslant 0 \tag{28}$$

(see, for example, [7]). Moreover, given a C_0 semigroup $\mathcal{T}$ on Y, its (*infinitesimal*) *generator* is the linear operator $A : D(A) \to Y$ defined by

$$Au = \lim_{t \searrow 0} \frac{1}{t}(T_t u - u)$$

on the domain $D(A) \subset Y$ composed of those $u \in Y$ for which the map $t \mapsto T_t u$ is differentiable on $\mathbb{R}^+$ (one can show that this is equivalent to the right-sided differentiability of the map $t \mapsto T_t u$ at 0). The generator A is closed, densely defined, and determines the semigroup uniquely (see [7]).

We continue to assume that $\|\cdot\|_t$, for $t \in \mathbb{R}$, is a family of norms on a Banach space X satisfying (16) for some continuous function R such that the map $(t, x) \mapsto \|x\|_t$ is continuous on $\mathbb{R} \times X$. Moreover, we continue to define C_* as in (25) in terms of the norms $\|\cdot\|_t$. Given a linear operator $B : C_* \to C_*$, we shall denote its operator norm by

$$\|B\|_* = \sup_{u \neq 0} \frac{\|Bu\|_*}{\|u\|_*}. \tag{29}$$

Now consider an evolution family $\mathcal{U}$ on X. For each $t \geqslant 0$, we define a linear operator $T_t : C_* \to C_*$ by

$$(T_t u)(s) = U(s, s-t)u(s-t) \quad \text{for } u \in C_*, s \in \mathbb{R}.$$

One can easily verify that $\mathcal{T} = (T_t)_{t \geqslant 0}$ is a semigroup on C_*, called the *evolution semigroup* associated with $\mathcal{U}$.

THEOREM 6. *Let $\mathcal{U}$ be an evolution family on a Banach space X. Then the following properties are equivalent:*

(1) *$\mathcal{U}$ is exponentially bounded with respect to the norms $\|\cdot\|_t$;*
(2) *$\mathcal{T}$ is a C_0 semigroup on C_*.*

Proof. (2 $\Rightarrow$ 1) If $\mathcal{T}$ is a C_0 semigroup on C_*, then by (28) we have

$$\|T_t\|_* = \sup_{u \neq 0} \frac{\sup_{r \in \mathbb{R}} \|(T_t u)(r)\|_r}{\sup_{s \in \mathbb{R}} \|u(s)\|_s} \leqslant M e^{\alpha t}$$

(see (29)) and so

$$\|U(r, r-t)u(r-t)\|_r = \|(T_t u)(r)\|_r \leqslant M e^{\alpha t} \sup_{s \in \mathbb{R}} \|u(s)\|_s. \tag{30}$$

Now take $x \in X$ and $t \in \mathbb{R}$. For each $n \in \mathbb{N}$, we consider a continuous function $\beta_{n,t} : \mathbb{R} \to [0, 1]$ such that

$$\beta_{n,t}(t) = 1 \quad \text{and} \quad \operatorname{supp} \beta_{n,t} \subset I_{n,t} := \left[t - \frac{1}{n}, t + \frac{1}{n}\right]. \tag{31}$$

For the function $u \in C_*$ defined by

$$u_n(s) = \beta_{n,t}(s)x \quad \text{for } s \in \mathbb{R}, \tag{32}$$

using (16) we have

$$\sup_{s \in \mathbb{R}} \|u_n(s)\|_s = \sup_{s \in \mathbb{R}} \|\beta_{n,t}(s)x\|_s = \sup_{s \in I_{n,t}} \|\beta_{n,t}(s)x\|_s \leqslant \sup_{s \in I_{n,t}} \|x\|_s. \tag{33}$$

By hypothesis, the map $s \mapsto \|x\|_s$ is continuous on $\mathbb{R}$ and so it is uniformly continuous on $I_{n,t}$. Therefore,

$$\lim_{n\to\infty} \sup_{s\in I_{n,t}} \|x\|_s = \|x\|_t. \tag{34}$$

By (30), we obtain

$$\|U(t+\tau,t)\beta_{n,t}(t)x\|_{t+\tau} = \|(T_\tau u_n)(t+\tau)\|_{t+\tau} \leqslant Me^{\alpha\tau} \sup_{s\in I_{n,t}} \|x\|_s$$

for $\tau \geqslant 0$. Finally, letting $n \to \infty$, it follows from (34) that

$$\|U(t+\tau,t)x\|_{t+\tau} \leqslant Me^{\alpha\tau}\|x\|_t.$$

This shows that $\mathcal{U}$ is exponentially bounded with respect to the norms $\|\cdot\|_t$.

($1 \Rightarrow 2$) Now assume that $\mathcal{U}$ is exponentially bounded with respect to the norms $\|\cdot\|_t$. Then

$$\|(T_t u)(s)\|_s = \|U(s,s-t)u(s-t)\|_s \leqslant Me^{\alpha t}\|u(s-t)\|_{s-t}$$

and so

$$\|T_t\|_* = \sup_{u\neq 0} \frac{\sup_{s\in\mathbb{R}} \|(T_t u)(s)\|_s}{\sup_{s\in\mathbb{R}} \|u(s)\|_s} \leqslant Me^{\alpha t}. \tag{35}$$

This shows that $\mathcal{T}$ is exponentially bounded. Moreover,

$$\limsup_{|s|\to\infty} \|(T_t u)(s)\|_s \leqslant Me^{\alpha t} \lim_{|s|\to\infty} \|u(s-t)\|_{s-t} = 0$$

and so the operators $T_t : C_* \to C_*$ are well defined.

In order to show that $\mathcal{T}$ is a C_0 semigroup, it remains to prove that $\|T_t u - u\|_* \to 0$ when $t \to 0$, for each $u \in C_*$. First observe that

$$\begin{aligned}
\|T_t u - u\|_* &= \sup_{s\in\mathbb{R}} \|(T_t u)(s+t) - u(s+t)\|_{s+t} \\
&= \sup_{s\in\mathbb{R}} \|U(s+t,s)u(s) - u(s+t)\|_{s+t} \\
&\leqslant \sup_{s\in\mathbb{R}} \|U(s+t,s)u(s) - u(s)\|_{s+t} + \sup_{s\in\mathbb{R}} \|u(s) - u(s+t)\|_{s+t}.
\end{aligned}$$

For $t < 1$ and $u \in C_c(X)$, say with $\operatorname{supp} u \subset [a,b]$, we have

$$\|T_t u - u\|_* \leqslant \sup_{s\in[a,b]} \|U(s+t,s)u(s) - u(s)\|_{s+t} + \sup_{s\in[a-1,b]} \|u(s) - u(s+t)\|_{s+t}. \tag{36}$$

Note that the functions

$$f(s,t) = \|U(s+t,s)u(s) - u(s)\|_{s+t}$$

and

$$g(s,t) = \|u(s) - u(s+t)\|_{s+t}$$

are continuous. Indeed, by hypothesis the map $F(t,x) = \|x\|_t$ is continuous and so it suffices to note that both

$$f(s,t) = F(s+t, U(s+t,s)u(s) - u(s))$$

and

$$g(s,t) = F(s+t, u(s) - u(s+t))$$

are compositions of continuous functions. Since f and g are uniformly continuous, respectively, on the compact sets $[a, b] \times [0, 1]$ and $[a - 1, b] \times [0, 1]$, and vanish for $t = 0$, it follows readily from (36) that $\|T_t u - u\|_* \to 0$ when $t \to 0$. For an arbitrary $u \in C_*$, take a sequence $(u_n)_{n\in\mathbb{N}} \subset C_c(X)$ such that $\|u_n - u\|_* \to 0$ when $n \to \infty$. We have

$$\begin{aligned}\|T_t u - u\|_* &\leqslant \|T_t(u - u_n)\|_* + \|T_t u_n - u_n\|_* + \|u_n - u\|_* \\ &\leqslant (Me^{\alpha t} + 1)\|u - u_n\|_* + \|T_t u_n - u_n\|_*,\end{aligned}$$

in view of (35). It follows from (36) with u replaced by u_n that

$$\limsup_{t\to 0} \|T_t u - u\|_* \leqslant (M + 1)\|u - u_n\|_*.$$

Finally, letting $n \to \infty$ we find that

$$\limsup_{t\to 0} \|T_t u - u\|_* = 0.$$

This completes the proof of the theorem. □

§4. *Characterization of hyperbolicity via semigroups.* In this section, we give a complete characterization of the nonuniform hyperbolicity (of a nonuniformly exponentially bounded) evolution family $\mathcal{U}$ in terms of the C_0 semigroup $\mathcal{T}$ associated with $\mathcal{U}$.

Let $\mathcal{U}$ be an evolution family on a Banach space X and let $\|\cdot\|_t$, for $t \in \mathbb{R}$, be a family of norms on X satisfying (16) for some continuous function R such that the map $(t, x) \mapsto \|x\|_t$ is continuous on $\mathbb{R} \times X$. By Theorem 6, if $\mathcal{U}$ is exponentially bounded with respect to the norms $\|\cdot\|_t$, then it has associated a C_0 semigroup $\mathcal{T}$ on C_*. The semigroup $\mathcal{T}$ is said to be *hyperbolic* if:

(1) there exists a projection $P : C_* \to C_*$ satisfying $PT_t = T_t P$ for $t \geqslant 0$;
(2) the map

$$T_t|_{Q(C_*)} : Q(C_*) \to Q(C_*),$$

where $Q = \mathrm{Id} - P$, is onto and invertible for each $t \geqslant 0$;
(3) there exist $\lambda, N > 0$ such that

$$\|T_t P u\|_* \leqslant Ne^{-\lambda t}\|u\|_* \quad \text{and} \quad \|S_t Q u\|_* \leqslant Ne^{-\lambda t}\|u\|_* \tag{37}$$

for $t \geqslant 0$ and $u \in C_*$, where $S_t = (T_t|_{Q(C_*)})^{-1}$.

Taking $t = 0$ in (37) we obtain $\|P\|_* \leqslant N$ and $\|Q\|_* \leqslant N$ (using the operator norm in (29)). Therefore, $P, Q \in \mathcal{L}(C_*)$.

The following theorem is our first main result. It shows that the nonuniform hyperbolicity of a nonuniformly exponentially bounded evolution family can be completely characterized in terms of its evolution semigroup.

THEOREM 7. *Let $\mathcal{U}$ be an evolution family on X that it is exponentially bounded with respect to the norms $\|\cdot\|_t$. Then $\mathcal{U}$ is hyperbolic with respect to the norms $\|\cdot\|_t$ if and only if $\mathcal{T}$ is hyperbolic.*

Proof. We first assume that $\mathcal{U}$ is hyperbolic with respect to the norms $\|\cdot\|_t$. We define a projection P on C_* by

$$(Pu)(s) = P_s u(s) \quad \text{for } u \in C_*,\ s \in \mathbb{R}.$$

It follows readily from Proposition 3 that the function Pu is continuous (since it is the composition of continuous functions). Moreover,

$$\|P_s u(s)\|_s \leqslant N\|u(s)\|_s \to 0 \quad \text{when } s \to \infty.$$

Finally, since

$$\|Pu\|_* = \sup_{s\in\mathbb{R}} \|(Pu)(s)\|_s = \sup_{s\in\mathbb{R}} \|P_s u(s)\|_s \leqslant N \sup_{s\in\mathbb{R}} \|u(s)\|_s = N\|u\|_*,$$

we have a well-defined linear operator $P : C_* \to C_*$. Now we show that $\mathcal{T}$ is hyperbolic. For $u \in C_*$ and $t \geqslant 0$, we have

$$\begin{aligned}\|T_t Pu\|_* &= \sup_{s\in\mathbb{R}} \|(T_t Pu)(s)\|_s = \sup_{s\in\mathbb{R}} \|U(s, s-t)P_{s-t}u(s-t)\|_s \\ &\leqslant Ne^{-\lambda t} \sup_{s\in\mathbb{R}} \|u(s-t)\|_{s-t} \leqslant Ne^{-\lambda t}\|u\|_*.\end{aligned}$$

Moreover, the operator $T_t|_{Q(C_*)} : Q(C_*) \to Q(C_*)$ is invertible. Indeed, for each $u \in C_*$, the equation $T_t v = Qu$ has the unique solution

$$v(s) = V(s, s+t)Q_{s+t}u(s+t) \quad \text{for } s \in \mathbb{R},$$

which belongs to C_* since

$$\begin{aligned}\|v\|_* &= \sup_{s\in\mathbb{R}} \|v(s)\|_s = \sup_{s\in\mathbb{R}} \|V(s, s+t)Q_{s+t}u(s+t)\|_s \\ &\leqslant Ne^{-\lambda t} \sup_{s\in\mathbb{R}} \|u(s+t)\|_{s+t} = Ne^{-\lambda t}\|u\|_*.\end{aligned} \tag{38}$$

Finally, letting $S_t = (T_t|_{Q(C_*)})^{-1}$ we have $v = S_t Qu$ and so it follows from (38) that

$$\|S_t Qu\|_* \leqslant Ne^{-\lambda t}\|u\|_* \quad \text{for } u \in C_*,\ t \geqslant 0.$$

This shows that the semigroup $\mathcal{T}$ is hyperbolic.

Now we assume that $\mathcal{T}$ is hyperbolic, with projection P. □

LEMMA 1. *We have* $(Pu)(s) = P_s u(s)$ *for some strongly continuous map* $\mathbb{R} \ni s \mapsto P_s \in \mathcal{L}(X)$.

Proof of the lemma. Since $\mathcal{T}$ is hyperbolic, we have

$$P(C_*) \subset \{u \in C_* : T_t u \to 0 \text{ when } t \to \infty\}. \tag{39}$$

Moreover, there exist $\lambda, N > 0$ such that

$$N^{-1}e^{\lambda t}\|Qu\|_* \leqslant \|T_t Qu\|_* \leqslant \|T_t u\|_* + Ne^{-\lambda t}\|Pu\|_*$$

for $u \in C_*$ and $t \geqslant 0$. This readily implies that if $T_t u \to 0$ when $t \to \infty$, then $Qu = 0$. Together with (39) this shows that

$$P(C_*) = \{u \in C_* : T_t u \to 0 \text{ when } t \to \infty\}.$$

Now observe that for any $\beta \in C_c(\mathbb{R})$ (where $C_c(\mathbb{R})$ denotes the set of all real-valued continuous functions with compact support), we have

$$\begin{aligned}(T_t(\beta Pu))(s) &= U(s, s-t)\beta(s-t)(Pu)(s-t) \\ &= \beta(s-t)(T_t Pu)(s-t)\end{aligned}$$

and so

$$\|T_t(\beta Pu)\|_* \leqslant \|\beta\|_\infty \|T_t Pu\|_* \to 0$$

when $t \to \infty$. Therefore, $\beta Pu \in P(C_*)$ for all $\beta \in C_c(\mathbb{R})$. Moreover, $\beta u = \beta Pu + \beta Qu$ and so

$$P(\beta u) = P(\beta Pu) + P(\beta Qu) = \beta Pu + P(\beta Qu). \tag{40}$$

We want to show that $P(\beta Qu) = 0$. Writing $\beta_t(s) = \beta(s+t)$, we obtain

$$\|P(\beta Qu)\|_* = \|P(\beta T_t S_t Qu)\|_* = \|P(T_t(\beta_t S_t Qu))\|_* = \|T_t P(\beta_t S_t Qu)\|_*, \tag{41}$$

because T_t commutes with P. Since $\mathcal{T}$ is hyperbolic, it follows from (41) that

$$\|P(\beta Qu)\|_* \leqslant Ne^{-\lambda t}\|\beta_t S_t Qu\|_* \leqslant N^2 e^{-2\lambda t}\|\beta\|_\infty \|u\|_*$$

for $t \geqslant 0$. Finally, letting $t \to \infty$ we obtain $P(\beta Qu) = 0$ and it follows from (40) that

$$P(\beta u) = \beta Pu \quad \text{for } \beta \in C_c(\mathbb{R}),\ u \in C_*.$$

Given $n \in \mathbb{N}$ and $t \in \mathbb{R}$, let $\beta_{n,t}$ be the function in (31). For each $u \in C_*$, we have

$$\|(Pu)(t)\|_t = \|(\beta_{n,t} Pu)(t)\|_t \leqslant \|\beta_{n,t} Pu\|_* = \|P(\beta_{n,t} u)\|_* \leqslant \|P\|_* \|\beta_{n,t} u\|_* \tag{42}$$

(using the operator norm in (29)). On the other hand,

$$\|\beta_{n,t} u\|_* = \sup_{s\in\mathbb{R}} \|(\beta_{n,t} u)(s)\|_s = \sup_{s\in I_{n,t}} \|(\beta_{n,t} u)(s)\|_s \leqslant \sup_{s\in I_{n,t}} \|u(s)\|_s.$$

Since the map

$$s \mapsto \|u(s)\|_s = F(s, u(s)) \quad \text{with } F(t,x) = \|x\|_t$$

is continuous (because it is the composition of continuous functions), letting $n \to \infty$ we obtain

$$\limsup_{n\to\infty} \|\beta_{n,t} u\|_* \leqslant \limsup_{n\to\infty} \sup_{s\in I_{n,t}} \|u(s)\|_s = \|u(t)\|_t. \tag{43}$$

In particular, if $u(t) = 0$, then

$$\|(Pu)(t)\|_t \leqslant \|P\|_* \limsup_{n\to\infty} \|\beta_{n,t} u\|_* \leqslant \|P\|_* \|u(t)\|_t = 0$$

and so $(Pu)(t) = 0$. Hence, one can define linear operators P_t on X by

$$P_t x = (Pu)(t) \quad \text{for any } u \in C_* \text{ with } u(t) = x.$$

Moreover, by (42) and (43),

$$\|P_t x\|_t = \|(Pu)(t)\|_t \leqslant \|P\|_* \|u(t)\|_t = \|P\|_* \|x\|_t \tag{44}$$

and so $P_t \in \mathcal{L}(X)$. Finally, we show that the map $t \mapsto P_t$ is strongly continuous. Given $x \in X$ and $t \in \mathbb{R}$, take $u \in C_*$ such that $u(s) = x$ for all s in some open neighborhood I of t. Since $P_s x = (Pu)(s)$ for all $s \in I$ and Pu is continuous, we conclude that the map $t \mapsto P_t x$ is continuous. This completes the proof of the lemma. □

Note that

$$(PT_t u)(s) = P_s(T_t u)(s) = P_s U(s, s-t)u(s-t)$$

and

$$(T_t Pu)(s) = U(s, s-t)(Pu)(s-t) = U(s, s-t)P_{s-t}u(s-t).$$

Since $PT_t = T_tP$ and $u(s-t)$ can be chosen arbitrarily, we obtain

$$P_sU(s, s-t) = U(s, s-t)P_{s-t},$$

thus yielding property (11).

Now we observe that since the map $T_t|_{Q(C_*)} : Q(C_*) \to Q(C_*)$ is onto and invertible, we have

$$\{Qu : u \in C_*\} = \{T_tQv : v \in C_*\} \quad \text{for } t \geqslant 0.$$

Hence, for each $s \in \mathbb{R}$ and $t \geqslant 0$, the operator $Q_s = \mathrm{Id} - P_s$ satisfies

$$Q_s(X) = \{(Qu)(s) : u \in C_*\} = \{(T_tQv)(s) : v \in C_*\} = (U(s, s-t)Q_{s-t})(X)$$

and so the map

$$U(s, s-t)|_{Q_{s-t}(X)} : Q_{s-t}(X) \to Q_s(X) \tag{45}$$

is onto. In order to prove that it is one-to-one, given $x \in X$, for each $n \in \mathbb{N}$ we consider the function $u_n \in C_*$ defined by

$$u_n(\tau) = \beta_{n,s-t}(\tau)x \quad \text{for } \tau \in \mathbb{R},$$

with $\beta_{n,s-t}$ as in (31) (with t replaced by $s-t$). Then

$$\begin{aligned} \|T_tQu_n\|_* &= \sup_{\tau\in\mathbb{R}} \|(T_tQu_n)(\tau)\|_\tau \\ &= \sup_{\tau\in\mathbb{R}} \|U(\tau, \tau-t)Q_{\tau-t}\beta_{n,s-t}(\tau-t)x\|_\tau \\ &\leqslant \sup_{\tau\in I_{n,s}} \|U(\tau, \tau-t)Q_{\tau-t}x\|_\tau. \end{aligned} \tag{46}$$

In view of Lemma 1 together with the continuity of the map $(t, x) \mapsto \|x\|_t$ and the strong continuity of the evolution family $\mathcal{U}$, we find that the map

$$\tau \mapsto \|U(\tau, \tau-t)Q_{\tau-t}x\|_\tau$$

is continuous and so uniformly continuous on $I_{n,s}$. Therefore, letting $n \to \infty$ in (46) yields the inequality

$$\limsup_{n\to\infty} \|T_tQu_n\|_* \leqslant \|U(s, s-t)Q_{s-t}x\|_s. \tag{47}$$

On the other hand, we have

$$\begin{aligned} Ne^{-\lambda t}\|T_tQu_n\|_* &\geqslant \|Qu_n\|_* \geqslant \|(Qu_n)(s-t)\|_{s-t} \\ &= \|Q_{s-t}\beta_{n,s-t}(s-t)x\|_{s-t} = \|Q_{s-t}x\|_{s-t}. \end{aligned}$$

Hence, it follows from (47) that

$$\|U(s, s-t)Q_{s-t}x\|_s \geqslant \frac{1}{N}e^{\lambda t}\|Q_{s-t}x\|_{s-t}. \tag{48}$$

This shows that the operator $U(s, s-t)|_{Q_{s-t}(X)}$ in (45) is one-to-one and thus it is invertible. Finally, replacing x by $V(s-t, s)Q_sx$ in (48) we conclude that

$$\|V(s-t, s)Q_sx\|_{s-t} \leqslant Ne^{-\lambda t}\|Q_sx\|_s$$

and it follows from (44) that

$$\|V(s-t, s)Q_sx\|_{s-t} \leqslant N(1 + \|P\|)e^{-\lambda t}\|x\|_s.$$

Finally, we prove the other inequality in the notion of a hyperbolic evolution family with respect to some norms. Note that

$$\|U(\tau + r, \tau)P_\tau u(\tau)\|_{\tau+r} = \|(T_r Pu)(\tau + r)\|_{\tau+r} \leqslant \|T_r Pu\|_* \leqslant Ne^{-\lambda r}\|u\|_* \tag{49}$$

for $r \geqslant 0$. Now take $x \in X$ and $t \in \mathbb{R}$. Given $n \in \mathbb{N}$ and $t \in \mathbb{R}$, we consider a continuous function $\beta_{n,t} : \mathbb{R} \to [0, 1]$ satisfying (31) and we define $u_n \in C_*$ by (32). In view of (33) and (49), we obtain

$$\|U(\tau + r, \tau)P_\tau \beta_{n,t}(\tau)x\|_{\tau+r} \leqslant Ne^{-\lambda r}\|u_n\|_* \leqslant Ne^{-\lambda r} \sup_{s \in I_{n,t}} \|x\|_s$$

for $r \geqslant 0$. Hence, taking $\tau = t$ and letting $n \to \infty$ yields the inequality

$$\|U(t + r, t)P_t x\|_{t+r} \leqslant Ne^{-\lambda r}\|x\|_t.$$

This concludes the proof of the theorem. □

§5. *Further characterizations of hyperbolicity.* In this section, we obtain further characterizations of the hyperbolicity of an evolution family in terms of its evolution semigroup.

We first establish a spectral mapping theorem for the evolution semigroup associated with an evolution family. Let $\|\cdot\|_t$, for $t \in \mathbb{R}$, be a family of norms on a Banach space X satisfying (16) for some continuous function R such that the map $(t, x) \mapsto \|x\|_t$ is continuous on $\mathbb{R} \times X$.

THEOREM 8. *Let $\mathcal{U}$ be an evolution family on X that it is exponentially bounded with respect to the norms $\|\cdot\|_t$ and let $\mathcal{T}$ be its evolution semigroup. Then for each $t > 0$, the spectrum $\sigma(T_t)$ is invariant under rotations centered at the origin. Moreover, the generator G of $\mathcal{T}$ satisfies*

$$\sigma(T_t) \setminus \{0\} = e^{t\sigma(G)} \quad \text{for } t \geqslant 0. \tag{50}$$

Proof. For each $\mu \in \mathbb{R}$ and $u \in C_*$, let

$$(M_\mu u)(s) = e^{i\mu s}u(s) \quad \text{for } s \in \mathbb{R}.$$

Since $\|(M_\mu u)(s)\|_s = \|u(s)\|_s$, we have $\|M_\mu u\|_* = \|u\|_*$. Hence, we obtain an isomorphism $M_\mu : C_* \to C_*$ of C_*. Moreover,

$$M_\mu T_t M_{-\mu} = e^{i\mu t}T_t.$$

Therefore,

$$\sigma(T_t) = \sigma(M_\mu T_t M_{-\mu}) = \sigma(e^{i\mu t}T_t) = e^{i\mu t}\sigma(T_t)$$

and the arbitrariness of μ yields the first statement in the theorem.

For the last property, we recall that the spectrum $\sigma(A)$ of a closed operator $A : D(A) \subset Y \to Y$ is given by

$$\sigma(A) = \sigma_a(A) \cup \sigma_r(A)$$

(the union may not be disjoint), where $\sigma_a(A)$ is the *approximate point spectrum* of A, composed of those $\lambda \in \mathbb{C}$ for which $A - \lambda\mathrm{Id}$ is not one-to-one or is not closed in Y, and $\sigma_r(A)$ is the *residual spectrum* of A, composed of those $\lambda \in \mathbb{C}$ for which $A - \lambda\mathrm{Id}$ is not dense in Y. For example, by [4, Theorems IV.3.6 and IV.3.7], we have

$$\sigma_a(T_t) \supset e^{t\sigma_a(G)} \quad \text{and} \quad \sigma_r(T_t) \setminus \{0\} = e^{t\sigma_r(G)}$$

for $t \geqslant 0$. Hence, in order to establish (50) it suffices to prove that

$$\sigma_a(T_t) \setminus \{0\} \subset e^{t\sigma_a(G)} \quad \text{for } t \geqslant 0.$$

This is clear for $t = 0$. Now take $t > 0$ and $\lambda \in \sigma_a(T_t) \setminus \{0\}$. For any $\mu \in \mathbb{C}$ such that $e^{i\mu t} = \lambda$, the family $(e^{-i\mu t}T_t)_{t\geqslant 0}$ is again an evolution semigroup and 1 is an eigenvalue of $e^{-i\mu t}T_t = (1/\lambda)T_t$. Hence, it suffices to show that if $1 \in \sigma_a(T_t)$ for some $t > 0$, then $0 \in \sigma_a(G)$.

Assume that $1 \in \sigma_a(T_t)$ for some $t > 0$. For example, by [4, Lemma IV.1.9], $0 \in \sigma_a(G)$ if and only if there exists a sequence $(v_n)_{n\in\mathbb{N}}$ in $D(G)$ with $\|v_n\|_* = 1$ for each n such that $\|Gv_n\|_* \to 0$ when $n \to \infty$. By the same lemma, there exists also a sequence $(u_n)_{n\in\mathbb{N}}$ in C_* such that

$$\|u_n\|_* = 1 \quad \text{and} \quad \|T_{kt}u_n - u_n\|_* < \tfrac{1}{2} \tag{51}$$

for $k = 0, 1, \ldots, 2n$. Indeed,

$$\|T_{kt}u_n - u_n\|_* \leqslant \sum_{j=0}^{k-1} \|T_{(j+1)t}u_n - T_{jt}u_n\|_* \leqslant \sum_{j=0}^{2n-1} \|T_t\|^j \|T_t u_n - u_n\|_*$$

and so it suffices to choose u_n such that $\|T_t u_n - u_n\|_*$ is sufficiently small. It follows from the inequalities

$$\|u_n\|_* - \|T_{kt}u_n - u_n\|_* \leqslant \|T_{kt}u_n\|_* \leqslant \|T_{kt}u_n - u_n\|_* + \|u_n\|_*$$

together with (51) that

$$\tfrac{1}{2} < \sup_{s\in\mathbb{R}} \|U(s, s-kt)u_n(s-kt)\|_s < \tfrac{3}{2} \tag{52}$$

for $k = 0, 1, \ldots, 2n$. In particular, given $n \in \mathbb{N}$, there exists $s_n \in \mathbb{R}$ such that

$$\|U(s_n, s_n - nt)x_n\|_{s_n} > \tfrac{1}{2} \quad \text{for } x_n := u_n(s_n - nt).$$

Now consider a C^1 function $\kappa_n : \mathbb{R} \to [0, 1]$ with compact support such that

$$\kappa_n(s_n) = 1, \quad \operatorname{supp}\kappa_n \subset I_n := [s_n - nt, s_n + nt], \quad \|\kappa_n'\|_\infty \leqslant \frac{2}{nt} \tag{53}$$

and let

$$v_n(s) = \kappa_n(s)U(s, s_n - nt)x_n \quad \text{for } s \in \mathbb{R}.$$

Clearly, $v_n \in C_*$ (since κ_n has compact support) and

$$\|v_n\|_* \geqslant \|v_n(s_n)\|_{s_n} > \tfrac{1}{2}. \tag{54}$$

Moreover,

$$\begin{aligned}(T_\tau v_n)(s) &= \kappa_n(s-t)U(s, s-\tau)U(s-\tau, s_n - n\tau)x_n \\ &= \kappa_n(s-\tau)U(s, s_n - nt)x_n.\end{aligned}$$

Thus, $v_n \in D(G)$ and

$$(Gv_n)(s) = \lim_{\tau\searrow 0} \frac{1}{\tau}(T_\tau v_n - v_n)(s) = -\kappa_n'(s)U(s, s_n - nt)x_n.$$

Now write $s \in I_n$ in the form

$$s = s_n + (k + r - n)t$$

for some $k = 0, 1, \ldots, 2n$ and $r \in [0, 1)$. Taking $t_n = s_n + (k-n)t$, it follows from (52) and (53) together with (15) that

$$\begin{aligned}\|(Gv_n)(s)\|_s &\leqslant \frac{2}{nt}\|U(s, s_n - nt)x_n\|_s \\ &\leqslant \frac{2}{nt}Me^{\alpha t}\|U(s_n + (k-n)t, s_n - nt)x_n\|_{s_n+(k-n)t} \\ &= \frac{2M}{nt}e^{\alpha t}\|U(t_n, t_n - kt)u_n(t_n - kt)\|_{t_n} \leqslant \frac{3M}{nt}e^{\alpha t}\end{aligned}$$

for $s \in I_n$. For the functions $\bar{v}_n = v_n/\|v_n\|_*$ in $D(G)$ we have $\|\bar{v}_n\|_* = 1$ and it follows from (54) that

$$\|G\bar{v}_n\|_* = \sup_{s\in I_n}\frac{\|(Gv_n)(s)\|_s}{\|v_n\|_*} \leqslant \frac{6M}{nt}e^{\alpha t} \to 0$$

when $n \to \infty$. Thus, $0 \in \sigma_a(G)$ and the proof of the theorem is complete. □

Finally, we formulate our second main result. Let $\mathcal{C} = \{z \in \mathbb{C} : |z| = 1\}$ and for a nonuniformly hyperbolic evolution family $\mathcal{U}$ let

$$\Gamma(t, s) = \begin{cases} U(t, s)P_s & \text{if } t \geqslant s, \\ -V(t, s)Q_s & \text{if } t < s. \end{cases}$$

THEOREM 9. *Let $\mathcal{U}$ be an evolution family on X that it is exponentially bounded with respect to the norms $\|\cdot\|_t$ and let $\mathcal{T}$ be its evolution semigroup. Then $\mathcal{T}$ is hyperbolic if and only if $\sigma(T_t) \subset \mathbb{C} \setminus \mathcal{C}$ for some $t > 0$ (and so for all $t > 0$). In this case, the generator G of $\mathcal{T}$ is invertible and*

$$(G^{-1}u)(s) = -\int_{\mathbb{R}}\Gamma(s, \tau)u(\tau)\,d\tau \quad \text{for } u \in C_*,\ s \in \mathbb{R}. \tag{55}$$

Proof. The first statement is a direct consequence of Theorem 8.

For the second statement, we note that in view of Theorem 8, the generator G is invertible if and only if $\sigma(T_t) \subset \mathbb{C} \setminus \mathcal{C}$. Now assume that G is invertible. First observe that since

$$\begin{aligned}\int_{\mathbb{R}}\|\Gamma(s, \tau)u(\tau)\|_s\,d\tau &\leqslant \int_{-\infty}^{s} Ne^{-\lambda(s-\tau)}\|Pu\|_*\,d\tau + \int_s^{\infty} Ne^{-\lambda(\tau-s)}\|Qu\|_*\,d\tau \\ &\leqslant \frac{2N}{\lambda}\|u\|_* < \infty,\end{aligned}$$

the integral in (55) converges. Denoting by $(\Phi u)(s)$ the right-hand side of (55), we obtain

$$\begin{aligned}(\mathrm{Id} - T_t)(\Phi u)(s) &= \int_{-\infty}^{s} U(s, \tau)P_\tau u(\tau)\,d\tau - \int_s^{\infty} V(s, \tau)Q_\tau u(\tau)\,d\tau \\ &\quad - U(s, s-t)\int_{-\infty}^{s-t} U(s-t, \tau)P_\tau u(\tau)\,d\tau \\ &\quad + U(s, s-t)\int_{s-t}^{\infty} V(s-t, \tau)Q_\tau u(\tau)\,d\tau[0]\end{aligned}$$

$$= \int_{s-t}^{s} U(s,\tau)P_\tau u(\tau)\,d\tau + \int_{s-t}^{s} U(s,\tau)Q_\tau u(\tau)\,d\tau$$

$$= \int_{s-t}^{s} U(s,\tau)u(\tau)\,d\tau = \int_0^t (T_\tau u)(s)\,d\tau.$$

Finally, dividing by $t > 0$ and letting $t \searrow 0$ yields the identity $-G\Phi u = u$. One can prove in a similar manner that for each $u \in D(G)$, $t > 0$ and $s \in \mathbb{R}$ we have

$$(\Phi(\mathrm{Id} - T_t)u)(s) = \int_0^t (T_\tau u)(s)\,d\tau.$$

Hence, proceeding as before, we obtain $-\Phi G u = u$ and so $G^{-1} = -\Phi$. □

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